Linking and Unlinking

How a node links to ASL3, DVSwitch and USRP2M17: the bridge nodes that carry each one, the commands each tool sends, the paths for a connect and a disconnect, and what is left behind when a link is dropped.

ASL3 DVSWITCH USRP2M17

SUMMARY · READ-ONLY REVIEW

Source: the tool files and configs on branch ccr-9d4fe19f-ssud83. dvswitch.sh is not in the repository; its commands are described from how the tools call it.

What is covered

Three kinds of link. ASL3 links nodes to each other through Asterisk. DVSwitch and USRP2M17 are reached through a bridge node that Asterisk links in, so a radio on the node can reach the bridge.

1.1 Which tools link

TOOL	ROLE IN LINKING
Node Controller	The main controller. Makes and drops all three kinds of link. Holds the Dashboard's own state.
Node Talker	Makes the same links. It hands them to Node Controller when Node Controller is running, and does them itself when it is not (the "solo" path).
Node Recorder	Carries a copy of Node Talker's DVSwitch and M17 functions (checked by name), with the same delegate and solo paths.
SysMon	Restarts the USRP2M17 service and refreshes its reflector list. It does not connect or disconnect.
44helper	Installs USRP2M17 and DVSwitch-related pieces. It does not connect or disconnect.

1.2 Sources and limits

- The Node Controller and Node Talker paths were read in full. Node Recorder's copy was checked for the same functions.
- `dvswitch.sh` lives outside the repository, at `/opt/MMDVM_Bridge/dvswitch.sh` or one of three other paths. What it does with each argument is taken from the calls the tools make, not from its source.
- The Asterisk link codes (1, 3, 6, 11) are described in the Connect and Disconnect Review. Their meanings come from how the code uses them.
- Nothing was run on a node. Behaviour is described from the code.

Three bridge nodes

Each kind of link goes through one bridge node. The bridge node is an ordinary Asterisk node, linked to the main node with the same link command as any other node.

NODE	CARRIES	AUDIO PATH (RPT.CONF RXCHANNEL)
1999, digital bridge	DVSwitch (DMR, D-STAR, YSF, P25, NXDN, STFU and others)	`USRP/127.0.0.1:34001:32001`
1917, M17 bridge	USRP2M17	`USRP/127.0.0.1:34008:32008`
1921, Node Talker's transmit node	The node Node Talker keys to send a person's voice. Node Recorder uses the same node (it is handed over in rpt.conf) and receives its audio through Node Talker's forwarding port. Not a bridge for a mode	`USRP/127.0.0.1:34018:32018`

The ports for node 1917 match the USRP2M17 ini (Part 5), so the bridge and the M17 program agree on where audio goes. Node 1999's ports go to the DVSwitch side of the bridge; the DVSwitch configuration is outside the repository, so the match is not checked here.

2.1 The three link types

KIND	CONNECTS	SHOWS ON THE NODE AS
ASL3	Another node, by its number (or an EchoLink node through the gateway)	A linked node in Asterisk. This is the normal AllStar link.
DVSwitch	The bridge node 1999 is linked to the main node, and the bridge is then tuned to a talkgroup, reflector or gateway	The bridge node linked, plus a tuned mode and target in the Dashboard
USRP2M17	The bridge node 1917 is linked, and the USRP2M17 program is pointed at a reflector	The bridge node linked, plus a reflector and module in the ini

Linking nodes in Asterisk

3.1 Commands

Every ASL3 link is one of these commands, sent through the Asterisk manager interface (or the command line as a fallback):

COMMAND	USED FOR
`rpt cmd <main> ilink 3 <node>`	Connect two-way (the usual connect, and linking a bridge node)
`rpt cmd <main> ilink 2 <node>`	Connect receive-only, for a phone node set to listen only
`rpt cmd <main> ilink 1 <node>`	Drop one link
`rpt cmd <main> ilink 11 <node>`	Drop a permanent link (one listed under `nodes=` in rpt.conf)
`rpt cmd <main> ilink 6`	Drop all links (used only when the link list cannot be read)

Link commands are sent to the main node, the one whose number is set in the Dashboard, and name the target as the second argument.

3.2 Rules that apply to every bridge

- **One link at a time.** Before a bridge or an ASL node is linked, every other link is dropped (up to 16 per pass). The Node Talker transmit node (1921) is kept.
- **Bridge slots are owned by the dashboard.** An ASL or EchoLink connect refuses a bridge node. The local API refuses to disconnect one.
- **Bridge reaper and watchdog.** In Node Controller, a bridge that is linked while another tab owns a different slot is dropped. A bridge the tab owns that falls out is linked again. Both wait out an 8-second cooldown.
- **Node Talker's guard** runs every 5 seconds, and only while Node Talker owns the node (not when Node Controller is running and the node is the Dashboard's). A condition must hold for two polls in a row, with an 8-second cooldown per link. It drops a bridge link that is not the bridge of what Node Talker has tuned. It re-links the bridge that Node Talker owns if that bridge falls out. When more than one public link is up (not the transmit node, not a bridge, and not a private node), it keeps one (the last ASL link Node Talker made, or else the first) and drops the others only if they are favorites; if none of the extras is a favorite, it logs that it left them alone. A link listed under `nodes=` in rpt.conf is a permanent link, and the guard drops it with `ilink 11` instead of `ilink 1`.

Linking to DVSwitch

4.1 The `dvswitch.sh` calls

CALL	MEANING AS USED BY THE TOOLS	TIMEOUT
`dvswitch.sh mode <MODE>`	Set the bridge's digital mode: DMR, DSTAR, YSF, P25, NXDN, STFU. FCS is sent as YSF.	40 s
`dvswitch.sh tune <value>`	Tune the current mode. For DMR this is a talkgroup number, or `password@address` to join a network. For D-STAR it is a reflector or gateway string.	40 s
`dvswitch.sh tune disconnect`	Leave the current talkgroup. Also sent as `*88` from the Dashboard's tune path.	10 s
`dvswitch.sh tune "  U"`	D-STAR unlink (seven spaces and a U)	10 s

Any other subcommand has a 10-second timeout. A call that times out is reported as "it may still finish", because the bridge can complete the step after the tool gives up.

4.2 D-STAR strings

A D-STAR tune string is the base callsign padded to six characters, then the module letter, then `L`. For example, `REF001 C` becomes `REF001 CL`. The Dashboard's XLX tune uses the same form. A gateway callsign has three to six characters. An entry with no module letter is refused.

4.3 DMR networks

A DMR favorite names a network. The tool joins it with a tune string `password@address`, and remembers the last one joined, so it does not join the same network again. A mode change clears that memory. When Node Controller has no DMR network for a favorite, Node Talker asks to add it, or to tune it once without saving it.

4.4 Node Controller path

1. Drop any other link, and clear a foreign link that is not the dashboard's own.
2. For D-STAR: send the D-STAR unlink and the disconnect, then wait 400 ms.
3. On ASL, link the digital bridge (1999) if it is not already linked (`ilink 3`).
4. Set the mode with `dvswitch.sh mode`, then wait out the settle time.
5. For a D-STAR gateway with ircDDB remote control set up, link through remote control and tune `CQCQCQ`. If that fails, fall back to a plain tune.
6. Tune the target. A timeout returns a message and does not update the page state.

The Dashboard keeps the state: the page, the favorite, the mode and the DMR server. Node Talker reads that state through `dvs-status`.

4.5 Node Talker and Node Recorder, delegate path

When Node Controller is running on the same node, Node Talker sends the request to it through the local API, with the local key in the request header:

- `dvs-tune`: mode, talkgroup, name, and for DMR the network. Node Controller refuses a D-STAR string with no module letter. For DMR it looks up the server by address or name. If the server is not known and the request is not "once", it refuses and asks for the network to be added.
- `dvs-disconnect`: disconnects the DVSwitch favorite Node Controller is tuned to.
- `dvs-status`: returns the page, the favorite, the server and its address, as JSON.
- `dmr-server-add`: adds a DMR network to Node Controller.

Node Talker sends the request only if the node is the Dashboard's node. A request for another node is refused with a message that says where to send that node's audio.

4.6 Node Talker solo path

With no Node Controller running, Node Talker does the same steps itself:

1. Take the tune lock (it waits up to 1 s), then drop every other link, except the bridge.
2. Arm the guard and leave the previous mode (`_leave`).
3. Start the gateway service for the mode if it is not running, and wait for it to become active.
4. Link the digital bridge with `ilink 3` if it is not linked.
5. Send `dvswitch.sh tune disconnect`, wait 0.4 s, then set the mode if it has changed.
6. For DMR, join the network if the last join was a different one. Wait 1 s.
7. Tune the target number, and save the state to the tuned-state file (mode, target, name, bridge).

4.7 Unlinking DVSwitch

PATH	WHAT IT DOES
Node Controller disconnect (D-STAR)	If ircDDB remote control is set up, sends an unlink. Then the D-STAR unlink and the talkgroup disconnect.
Node Controller disconnect (other modes)	The talkgroup disconnect. The bridge stays linked to the node.
Node Talker solo disconnect	`_leave`: for D-STAR, the D-STAR unlink; then the talkgroup disconnect; then `ilink 1` on the bridge. The state is cleared.
Node Talker, when Node Controller takes over	The state is cleared without a command, once Node Controller is seen running.

Notice that the bridge is unlinked from the node in the solo path, but not in the Node Controller path for other modes. A link left up in that case is shown as the bridge node still linked (Part 6).

Linking to M17

5.1 Files and service

ITEM	WHAT IT IS
`/opt/USRP2M17/USRP2M17.ini`	The program's config. Holds the callsign, the reflector name and address, the module, and the ports.
`usrp2m17.service`	The systemd unit. It must be restarted to read a new ini.
`/var/lib/mmdvm/M17Hosts.json`	The reflector list the tools use to look up an address for a reflector name. SysMon refreshes it.
Bridge node 1917	The Asterisk node the program's audio goes through (Part 2).

5.2 The ini that is written

Every connect and disconnect writes the whole file from a template. The fields that change are the callsign, the reflector name with its module, and the address. Node Controller and Node Talker use the same template:

SECTION	FIELD	VALUE
M17 Network	Callsign	The node's callsign
M17 Network	Address, Name	The reflector address, and "reflector module" (for example `M17-USA C`)
M17 Network	LocalPort, DstPort	17010 and 17000
M17 Network	GainAdjustdB, Daemon, Debug	3, 1, 0
USRP Network	Address, DstPort, LocalPort	127.0.0.1, 32008, 34008
Log	FilePath, FileRoot	`/var/log/usrp`, `USRP2M17`

The disconnected state is the same template with the name `DISCONNECTED`, the address `0.0.0.0` and module `A`.

5.3 Connect

Node Controller (`action_m17_connect`): check the reflector address, the base name and the module (A to Z, or `@ALL`), and that the M17 bridge is present. Clear any foreign link. Send the talkgroup disconnect. Write the ini, restart the service, link node 1917 with `ilink 3` if ASL is up and it is not linked, and set the page to M17 with the favorite `base|module`.

Node Talker, solo (`solo_m17`): check that USRP2M17 is present and that a callsign is set. Look up the address in M17Hosts.json, or use the favorite's own address. Take the lock, drop other links, leave the previous mode, and send the DVSwitch disconnect if DVSwitch is present. Write the ini, restart the service, and link node 1917. Save the state.

Delegate: Node Talker sends `m17-connect`. Node Controller looks the reflector up in its own list or in the M17 favorites, checks that the module is one the reflector publishes, and then runs the same connect as the Dashboard.

5.4 Disconnect

PATH	WHAT IT DOES
Node Controller, `m17-disconnect`	Unlinks node 1917 with `ilink 1` if ASL is up. Writes the disconnected ini and restarts the service. If the ini write fails, the message says the link was dropped but the file was not written.
Node Talker, solo	Park the ini (disconnected template) and restart the service, then `ilink 1` on node 1917.
Node Talker, delegate	Sends `m17-disconnect` to Node Controller, and only if an M17 reflector is set there.

5.5 SysMon and 44helper

- **SysMon** has an M17 tab with a Restart button (restarts `usrp2m17`, root only) and an Update reflector list button, which fetches M17Hosts.json. It also shows the ini and a sample stanza, and has an editor for the ini. The M17 tab does not tune or link.
- **44helper** has an M17 install tab. It clones and runs the USRP2M17-Selector installer, builds the program, and sets a sudoers rule that lets the web server restart the service. Its purge steps stop and remove the service. It does not link.

What is left behind

A disconnect has to drop the audio path, the bridge's mode or reflector, and the saved state. The table shows what each path does for each part.

PATH	BRIDGE NODE	MODE OR REFLECTOR	SAVED STATE
Node Controller, ASL disconnect	Not a bridge node, so no change	Not applicable	Cleared
Node Controller, DVSwitch disconnect	Stays linked (other modes)	Talkgroup disconnected	Page state cleared
Node Controller, D-STAR disconnect	Stays linked	D-STAR unlinked, then talkgroup disconnected	Page state set to Ready
Node Controller, M17 disconnect	`ilink 1` on 1917 (if ASL)	ini parked, service restarted	Page state cleared
Node Talker solo, DVSwitch	`ilink 1` on 1999	D-STAR unlink (if D-STAR), talkgroup disconnected	Tuned-state file cleared
Node Talker solo, M17	`ilink 1` on 1917	ini parked, service restarted	Tuned-state file cleared
Emergency reset (Node Controller)	Dropped with every other link	D-STAR unlink and talkgroup disconnect, then services restarted	Cleared

6.1 Two things a disconnect does not do

- A DVSwitch disconnect on the Node Controller path does not drop the bridge node. The bridge stays linked and the talkgroup is left disconnected.
- An M17 disconnect does not change the mode of the digital bridge, and a DVSwitch disconnect does not change the reflector in the M17 ini.

Things the code shows

7.1 Two implementations of the same link

Node Controller, Node Talker and Node Recorder each carry DVSwitch and M17 linking code. The M17 ini template is the same in all three, checked line by line for Node Talker and Node Recorder, and Node Controller's disconnect placeholder matches Node Talker's park. That match is by construction, not by a shared function, so a change to one has to be copied to the others. The book should say which tool a user should use for each job.

7.2 A failed tune can leave the bridge linked

In the Node Talker solo DVSwitch path, the bridge is linked before the mode is set and the frequency is tuned. If the tune then fails, the function returns an error. It does not drop the bridge or update the tuned-state file. The node is left with the bridge linked and no record of a target. The Node Controller path has the same ordering, but its failure messages do not say the bridge is still up.

7.3 A failed M17 restart leaves a new ini

In the Node Talker solo M17 path, the ini is written before the service is restarted. If the restart fails, the function returns an error with the new ini on disk, and the bridge is not linked. The next restart of the service will read the new reflector.

7.4 Every M17 change restarts the service

Connecting, disconnecting and parking all restart `usrp2m17`. Each restart interrupts any M17 audio that is running, so a change of reflector or module is never seamless. The book should tell users this.

7.5 D-STAR unlink depends on exact spacing

The D-STAR unlink is the string seven spaces and a `U`, sent exactly. The code sends it unchanged in every tool. A change in the spacing would break the unlink without any error from the code, so the string should not be reformatted.

7.6 Version gates

The README says Node Controller must be 9.3.94 or later for Node Talker to tune DVSwitch through it, and 9.3.89 or later for ASL and M17. Node Talker checks for an old version when it asks Node Controller to do a job, and reports that the version is too old. The version numbers in the README were not checked against the code for this summary.

7.7 Not checked

These were not traced for this summary: what `dvswitch.sh` does internally, how the bridge reports its state back, the ircDDB remote-control details, and the individual private-node numbers a home network uses. In the code, a "private" node is any all-digit node number up to 1999, and both the guard and the clean-up that runs before a connect leave those links alone.